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|---------------|--------------------------------|--------------------|---------------------------|
| Course Code   | CSC -111                       | Faculty            | Dr. Wazir Ali             |
| Course Title  | Object Oriented Programming    | Room# & Phone/Ext  | 202, FEST                 |
| Credit Hours  | 3+1                            | Consultation Hours | 9-5 PM                    |
| Pre-Requisite | CSC111 Programming Fundamental | E-Mail             | wazir.faculty@aror.edu.pk |
| Program       | BS (Cyber Security, AI, MMG)   | Semester           | Spring 2025               |

### 1. Course Objective

This course provides the introduction to object-oriented programming. The course aims to develop in students, Object Oriented Approach in programming and problem solving. The course targets basic Object-Oriented Programming principles, OOP process, basic OOP structure like Classes, Inheritance, Abstraction, Encapsulation, Composition etc. The course also aims to use JAVA language to demonstrate basic programming structure and Object-Oriented driven programming.

### 2. Course Description

This course introduces the concepts of object-oriented programming to students. The course begins with a brief review of control structures and data types with an emphasis on structured data types and array processing. It then moves on to introduce the object-oriented programming paradigm, focusing on the definition and use of classes along with the fundamentals of object-oriented design. Further topics, such as function overloading, operator overloading, composition, aggregation, inheritance, multiple inheritance, polymorphism, abstract classes and interfaces will also be covered.

### 3. Course Learning Outcomes:

| CO No. | Course Learning Outcomes (CLOs)                                                                                          | Graduate Attributes                                | Bloom's Taxonomy      |
|--------|--------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|-----------------------|
| 1      | <b>Interpret</b> an understanding of object-oriented paradigm and their mapping to object-oriented programming language. | GA_2<br>(Knowledge for Solving Computing Problems) | C2<br>(Understanding) |
| 2      | <b>Analyze</b> problems using object oriented programming constructs.                                                    | GA_3<br>(Problem Analysis)                         | C4<br>(Analyzing)     |

|   |                                                                           |                           |                  |
|---|---------------------------------------------------------------------------|---------------------------|------------------|
| 3 | <b>Apply</b> core object-oriented programming knowledge to solve problems | GA_3<br>(Problem Analysis | C3<br>(Applying) |
|   |                                                                           |                           |                  |

#### 4. Complex Computing Problem

|                                          |                                                                                                                                                                                                                                                                                |
|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Complex Computing Problem Details</b> | <p><b>Included: Yes</b><br/>Nature and details of Complex Computing Problem (CCP): It will be given in Assignment # 3.</p> <p>CCP will be based on CLO-3</p> <p><b>Attributes could be: CCP4</b><br/>CCP4: Depth of knowledge required</p> <p>Assessment in: Assignment# 3</p> |
|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

#### 5. Recommended Books

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| <b>TEXT BOOK:</b>                                                                                      |
| Schildt, Herbert, Java: The Complete Reference, 11 <sup>th</sup> Edition, McGraw-Hill, December , 2018 |
| <b>REFERENCE BOOKS:</b>                                                                                |
| Deitel, Paul, Java How to Program, 11th Edition, Pearson, 2017                                         |

#### 6. TEACHING PLAN:

| Week No. | Week Dates                 | Topics                                                                                                                                                                                                                     | Required Reading                                             | Key Date             |
|----------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|----------------------|
| 1.       | January 2026               | <b>Overview of the Course</b><br>Object Oriented Concepts, Procedural and Object Oriented, languages, Introduction to Basic Java Environment, Java Folder, Compilation and Running process, Java basic constructs,         | <b>Text Book:</b><br>Chapter 1 and Chapter 2<br><b>Pg: 3</b> |                      |
| 2.       | January 2026               | <b>Data Type, Variables and Arrays</b><br>Primitive data types, Defining arrays, Initializing, Accessing and Reusing Array Variables Array length, Multi dimension Arrays, Introducing string class and important methods. | <b>Text Book:</b><br>Chapter 3<br><b>Pg: 39</b>              |                      |
| 3.       | January 2026               | <b>Classes and Object</b><br>Class fundamental, Simple Class, Declaring Object, Use of new Object reference                                                                                                                | <b>Text Book:</b><br>Chapter 6<br><b>Pg: 117</b>             |                      |
| 4.       | January 2026               | <b>Methods and Constructor</b><br>Defining methods, defining constructors, parameters in a method, returning a value from method, use of this keyword                                                                      | <b>Text Book:</b><br>Chapter 6<br><b>Pg: 117</b>             | <b>Assignment #1</b> |
| 5.       | February 2026              | <b>Polymorphism</b><br>Overloading methods, Using objects as methods, Access controls                                                                                                                                      | <b>Text Book:</b><br>Chapter 7<br><b>Pg: 137</b>             | <b>Quiz # 1</b>      |
| 6.       | February 2026              | <b>Types of Polymorphism</b><br>Run time polymorphism, compile time polymorphism                                                                                                                                           | <b>Text Book:</b><br>Chapter 7<br><b>Pg: 137</b>             |                      |
| 7.       | <b>Midterm Examination</b> |                                                                                                                                                                                                                            |                                                              |                      |

|                          |               |                                                                                                                                                                                                    |                                                    |                       |
|--------------------------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|-----------------------|
| 8.                       | February 2026 | <b>Inheritance</b><br>Advantages of Inheritance, use of extend, use of super, Inheritance Hierarchies, Implementing Subclasses, The instance of Operator                                           | <b>Text Book:</b><br>Chapter 8<br><b>Pg: 171</b>   |                       |
| 9.                       | March 2026    | <b>Inheritance Types</b><br>Single, Multiple, Hierarchical, Multi-level, Hybrid                                                                                                                    | <b>Text Book:</b><br>Chapter 8<br><b>Pg: 171</b>   |                       |
| 10.                      | March 2026    | <b>Packages</b><br>Defining a Package, Importing Packages                                                                                                                                          | <b>Text Book:</b><br>Chapter 9<br><b>Pg: 199</b>   | <b>Assignment # 2</b> |
| 11                       | March 2026    | <b>Abstract Classes and Interfaces</b><br>Abstract Classes and Methods, Creating and Using Interfaces, Nested Interfaces, Final Methods and Classes, Static Methods in Interfaces, Default Methods | <b>Text Book:</b><br>Chapter 9<br><b>Pg: 199</b>   | <b>Quiz # 2</b>       |
| 12                       | March 2026    | <b>Exception Handling</b><br>Throwing Exceptions, Catching Exceptions, Checked and unchecked Exceptions, Designing Your Own Exception Types , Assertions , The try/finally Statement               | <b>Text Book:</b><br>Chapter 10<br><b>Pg: 227</b>  |                       |
| 13                       | April 2026    | <b>Enumeration in Java</b><br>Java enum, Java enum constructor, enum string.                                                                                                                       | <b>Text Book:</b><br>Chapter 12<br><b>Pg: 277</b>  | <b>Assignment # 3</b> |
| 14                       | April 2026    | <b>Swings</b><br>Swings features, components and containers, Event handling                                                                                                                        | <b>Text Book:</b><br>Chapter 32<br><b>Pg: 1061</b> |                       |
| 15                       | April 2026    | Object Oriented Design with UML                                                                                                                                                                    | <b>Handouts</b>                                    | <b>Quiz#3</b>         |
| 16                       | April 2026    | Assignment # 3 and Viva                                                                                                                                                                            |                                                    |                       |
| <b>Final Examination</b> |               |                                                                                                                                                                                                    |                                                    |                       |

## 7. RELATIONSHIP BETWEEN ASSESSMENT TOOLS AND CLOs:

| Assessment Tools    | CLO-1<br>(38 Marks) | CLO-2<br>(31 Marks) | CLO-3<br>(31 Marks) |
|---------------------|---------------------|---------------------|---------------------|
| <b>Quizzes</b>      | 7.8% (3)            | 19.3% (6)           | 19.3% (6)           |
| <b>Assignments</b>  | 13.1% (5)           | 16.1% (5)           | 16.1% (5)           |
| <b>Midterm Exam</b> | 52.6% (20)          | 32.3% (10)          | -                   |
| <b>Final Exam</b>   | 26.3% (10)          | 32.3% (10)          | 64.5% (20)          |

## 8. ASSESSMENT TOOLS & MARKS DISTRIBUTION:

| Assessment Tools | Percentage |
|------------------|------------|
| Quizzes          | 15%        |
| Assignments      | 15%        |
| Midterm Exam     | 30%        |
| Final Exam       | 40%        |

|              |             |
|--------------|-------------|
| <b>TOTAL</b> | <b>100%</b> |
|--------------|-------------|